

JCP campaign notebook

Continuous coordinate-network bases for Petrov–Galerkin reduced models

Tunable-NM-ROM project

Rebuilt October 6, 2026

What this is. The working notebook for the JCP paper: one entry per update, newest last, built by `notebook/build.sh`. It records decisions, designs, results (generated, never typed), and anything retracted. The canonical chronology stays in `LAB-LOG.md`; the plan is `reports/2026-10-06-jcp-offmesh-paper-plan.md`; the outline is `reports/2026-10-06-jcp-combined-paper-outline/`.

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Thesis

A reduced trial basis that is a *continuous, importance-ordered* function of position, $u(x) = G(x)c$, gives two properties. **Ordered:** one trained model offers an accuracy/cost dial at deployment (width R' , head on or off). **Continuous:** the nonlinear term is an integral, so it can be hyper-reduced by classical quadrature (tensor Gauss, rank-1 lattices) at points that ignore the mesh, instead of quadrature fitted to snapshots on the mesh. The paper combines the discarded ICLR material (the dial) with Hari's off-mesh quadrature study. The ICLR paper will not be submitted.

Decisions (user, 2026-10-06)

topic	decision
Paper	One combined JCP paper; ICLR dropped.
Off-mesh code	Copied byte-identically into each lane's <code>vendor/</code> with <code>PROVENANCE.json</code> (not forked from the quadrature branches).
Full-order comparators	Run <i>and compare against</i> all three of coarse-grid, second-order and spectral solvers, in addition to Newton-BiCGStab/GMRES and IMEX-CG. Supersedes the 2026-09-21 “named iterative only” rule.
Examples	Keep the current PDEs and add harder ones (chosen in the lane designs).
NS 3D	Keep the lane and keep improving its coordinate-network bank.
Worktrees	Claude names and creates them without asking (now in <code>CLAUDE.md</code> and <code>AGENTS.md</code>).
Notebook	This LaTeX notebook, rebuilt after every update.

Phase 0 checklist

- [\[done\]](#) Plan and five planning notes, revised after a Codex audit: `reports/2026-10-06-jcp-offmesh-paper-plan.md`.
- [\[done\]](#) Paper outline (PDF): `reports/2026-10-06-jcp-combined-paper-outline/`.
- [\[done\]](#) Literature check of the unread preprints and a 2024–2026 sweep (plan appendix 07); see below.
- [\[done\]](#) Week-1 lanes created (table below).
- [\[done\]](#) `CLAUDE.md`, `AGENTS.md` and memory updated with the decisions.
- [\[user\]](#) Withdraw the ICLR submission on OpenReview if it is still active.
- [\[user\]](#) Agree authorship and the split of work with Hari; get his consent to use his code and results.

Literature check: outcome

None of arXiv 2505.14595 (Φ -ROM, a grid-based contrast), 2507.07830 (meshless SPH projection, no hyper-reduction), 2508.21279 (energy-conservative EQP, a fitted method) or the LiCROM cubature (random mesh vertices, equal weights) pre-empts the claims. No ROM paper found uses weighted classical quadrature for the reduced residual. Per contribution:

- ordered tunable bank: **novel for ROMs**; credit nested dropout (Rippel et al. 2014), slimmable networks, POD truncation, LiCROM;
- classical quadrature replacing fitted hyper-reduction: **novel as a combination**; ingredients partly pre-empted (Monte Carlo/uniform points in Simplicits, SNF-ROM, Weder et al.; lattices in PINN training; fitted off-mesh points in CECM);
- decoupling from the mesh’s $O(h)$ error: **partly pre-empted** (SNF-ROM has the mechanism); Ingimarson–Rebholz–Iliescu (CMAME 2022) argue the opposite position and must be addressed;
- rule-family behaviour on tensor sine tests: **novel in the ROM setting**, as known theory with a new consequence;
- data-free error indicator: **partly pre-empted in principle**; call it an indicator, never a certificate.

Still to read before submission: arXiv 2607.27471, 2607.10965, 2602.01397, 2504.15657.

Week-1 lanes

All fork from `exp/2026-10-01-ns3d-coordnet-bank @ 21c175a1b`; each mirrors to `origin/codeonly/exp/<lane>` with its own `sync.github.sh`, and vendors 69 files of off-mesh code, each checked blob by blob against its source commit (`exp/2026-10-01-quadrature-study @ 84d28e303`, `exp/2026-10-01-quadrature-burgers3d @ aab6b6927`).

lane (<code>worktrees/</code> , branch <code>exp/...</code>)	experiments	role
2026-10-06-jcp-references	E0	order-verified second-order references; pseudo-spectral cross-check; sealed and out-of-distribution cohorts in a read-only store
2026-10-06-jcp-mechanism	E1, E2	rule ladders against theory; second-order stencil tensor, mesh-node analytic gradient, coarse-trained bank, all-continuum arm
2026-10-06-jcp-fom-pareto	E4	full-order cost ladders (iterative, IMEX-CG, second-order, coarse-grid, spectral); timing harness
2026-10-06-jcp-hr-baselines	E5	POD-LSPG/Galerkin with ECSW, Q-DEIM, GNAT; POD + exact tensor; best-practice EQ; CROM-style sampler

Harder examples: candidates for the designs

- time-dependent reaction–diffusion with e^u (Frank-Kamenetskii): non-polynomial, so no tensor exists;
- porous-medium diffusion $\nabla \cdot (u^m \nabla u)$ with non-integer m ;
- Burgers at lower viscosity / narrower bumps, to find where the rule size and bank floor break;
- a non-box domain (e.g. L-shape with a distance-function Dirichlet factor and a composite rule);
- NS 3D with an improved coordinate-network bank.

Carried-forward corrections (from the Codex audit of the plan)

Retracted: the explicit Gauss strip-width bound in planning note 03; “lattices converge like m^{-3} ” as an asymptotic rate (only a conditional upper bound); Hari’s “super-algebraic lattice” and “tent adds a kink” statements for this integrand; the per-lane sealing of test cohorts (now global); the 5 % timing fault under a 10 % gate; Gauss “exactness” on sines; the rule-doubling “certificate”; the 5–7-day schedule (at least 7.6 days on four GPUs).

Next (Phase 1)

For each week-1 lane: write `DESIGN.md` → Codex design audit → code → Codex code audit → real-size smoke with every control firing → cluster jobs → Codex results audit → generated report and a notebook entry. Lanes run as separate subagents, one worktree each, with distinct cluster namespaces.